Experiment 4

AIM: Use npm to install and manage packages

Step 1: Download from https://git-scm.com

Step 2: Akshar@v19 MINGW64 ~ (main)$ node -v

```
v22.13.1
```

Akshar@v19 MINGW64 ~ (main)$ npm -v

```
10.9.2
```

Akshar@v19 MINGW64 /d $mkdir npm-lab

Akshar@v19 MINGW64 /d $ cd npm-lab

Akshar@v19 MINGW64 /d/npm-lab $ npm init -y

```
Wrote to D:\npm-lab\package.json:

{
  "name": "npm-lab",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

Step 3: Install Required Package

Akshar@v19 MINGW64 /d/npm-lab $npm install express

```
added 65 packages, and audited 66 packages in 57s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Akshar@v19 MINGW64 /d/npm-lab$ npm install axios

```
added 11 packages, and audited 77 packages in 3s

24 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Akshar@v19 MINGW64 /d/npm-lab$ npm install nodemon --save-dev

```
added 26 packages, and audited 103 packages in 4s

29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Step 4:Create Web Server

```
Create a file named index.js and Add below content :
const express = require("express");
const axios = require("axios");
const app = express();
```

```
app.get("/", (req, res) => {
res.send("Hello Open Source Lab");
});

// JSON Route
app.get("/api/info", (req, res) => {
res.json({
name: "NPM Lab",
subject: "Open Source Technology",
semester: "BE 6th Sem"
});
});

// Axios Demo Route
app.get("/api/user", async (req, res) => {
const response = await axios.get("https://jsonplaceholder.typicode.com/users/1");
res.json(response.data);
});

app.listen(3000, () => {
console.log("Server running at http://localhost:3000");
});
```

Step 5: Run Server (Normal Way)

```
node index.js
Open browser:
http://localhost:3000
```

```
http://localhost:3000/api/info

http://localhost:3000/api/user
```

Step 6: Add npm Scripts

```
Open package.json and modify scripts section:


"scripts": {

"start": "node index.js",

"dev": "nodemon index.js"

}
```

Step 7: Run Using npm Script

```
1. Normal mode:

npm run start

2. Development mode (nodemon):


npm run dev

Now modify index.js text and save → Server automatically restarts (nodemon demo).
```

Step 8: Delete node_modules and Reinstall

```
Delete manually

Now reinstall:

npm install

node_modules folder recreated

Dependencies installed from package.json
```

Step 9: Add .gitignore File

PS D:\npm-lab> notepad .gitignore (open notepad write this after that save and close)

node_modules/

Step 10:Upload Project to Git (Without node_modules)

Akshar@v19 MINGW64 /d/npm-lab$ git init

```
Initialized empty Git repository in D:/npm-lab/.git/
```

Akshar@v19 MINGW64 /d/npm-lab (master)$ git config --global user.name "Vatsalkakkad125750-source"

Akshar@v19 MINGW64 /d/npm-lab (master)$ git config --global user.email"Vatsal.kakkad125750@marwadiuniversity.ac.in"

Akshar@v19 MINGW64 /d/npm-lab (master)$ git add .

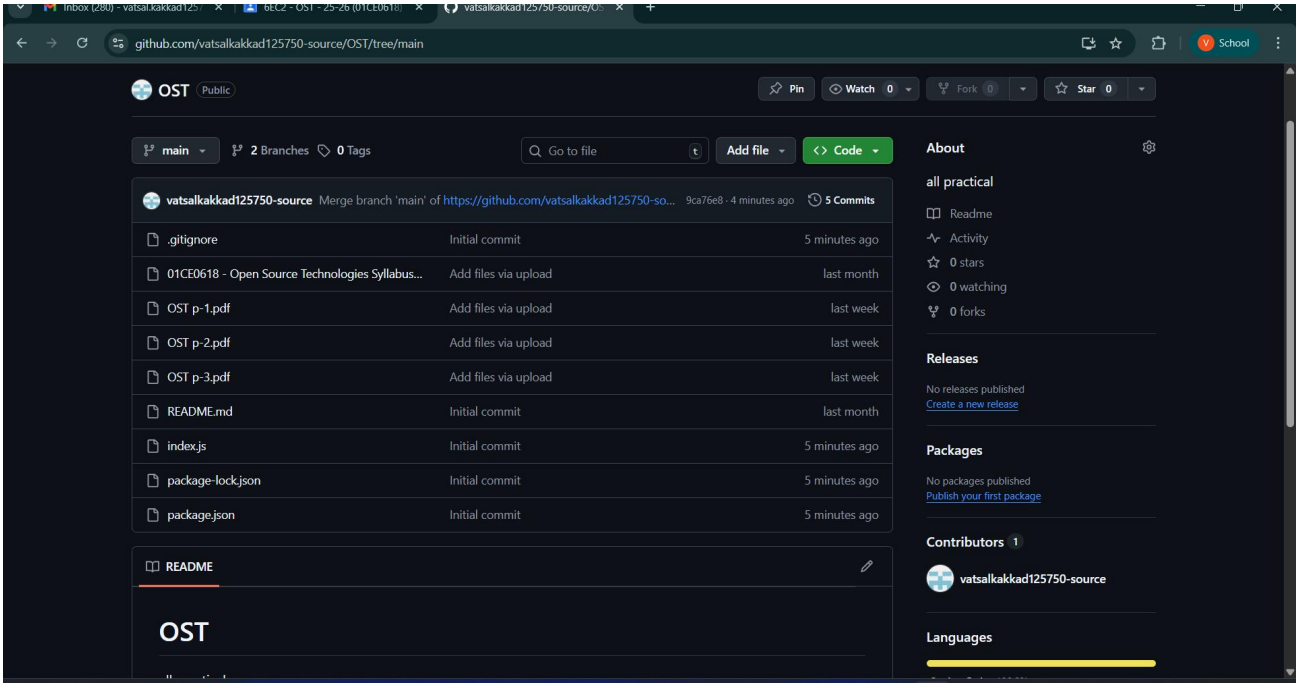
git commit -m "Initial commit"

git branch -M main

git push -u origin main

```
From https://github.com/vatsalkakkad125750-source/OST
 * branch            main       -> FETCH_HEAD
Already up to date.
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 12.50 KiB | 12.50 MiB/s, done.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/vatsalkakkad125750-source/OST.git
   749ef93..9ca76e8  main -> main
branch 'main' set up to track 'origin/main'.
```

Output:



```
are terminated then try again. if it still fails, a git process
may have crashed in this repository earlier:
remove the file manually to continue.
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/vatsalkakkad125750-source
/OST.git'

Akshardv19 MINGW64 /d/npm-lab (main)
$ ^C

Akshardv19 MINGW64 /d/npm-lab (main)
$ rm -f .git/index.lock

Akshardv19 MINGW64 /d/npm-lab (main)
$ ls .git
HEAD  config  description  hooks/  info/  objects/  refs/

Akshardv19 MINGW64 /d/npm-lab (main)
$ ls status
ls: cannot access 'status': No such file or directory

Akshardv19 MINGW64 /d/npm-lab (main)
$ git status
on branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        index.js
        package-lock.json
        package.json

nothing added to commit but untracked files present (use "git add" to track)

Akshardv19 MINGW64 /d/npm-lab (main)
$ git add .
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF
 the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the
next time Git touches it

Akshardv19 MINGW64 /d/npm-lab (main)
$ git commit -m "initial commit"
[main (root-commit) 27af87] initial commit
 4 files changed, 1359 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 index.js
 create mode 100644 package-lock.json
 create mode 100644 package.json

Akshardv19 MINGW64 /d/npm-lab (main)
$ git push -u origin main



remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 11 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (11/11), 1.35 MiB | 895.00 KiB/s, done.
From https://github.com/vatsalkakkad125750-source/OST
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Merge made by the 'ort' strategy.
 01CE0618 - Open Source Technologies Syllabus.pdf | Bin 0 -> 197419 bytes
 OST p-1.pdf                                      | Bin 0 -> 472076 bytes
 OST p-2.pdf                                      | Bin 0 -> 401515 bytes
 OST p-3.pdf                                      | Bin 0 -> 440037 bytes
 README.md                                        |   2 ++
 5 files changed, 2 insertions(+)
 create mode 100644 01CE0618 - Open Source Technologies Syllabus.pdf
 create mode 100644 OST p-1.pdf
 create mode 100644 OST p-2.pdf
 create mode 100644 OST p-3.pdf
 create mode 100644 README.md

Akshardv19 MINGW64 /d/npm-lab (main)
$ git pull origin main --allow-unrelated-histories
git push -u origin main
From https://github.com/vatsalkakkad125750-source/OST
 * branch            main       -> FETCH_HEAD
Already up to date.
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 12.50 KiB | 12.50 MiB/s, done.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/vatsalkakkad125750-source/OST.git
   749ef93..9ca76e8  main -> main
branch 'main' set up to track 'origin/main'.

Akshardv19 MINGW64 /d/npm-lab (main)
```